Al-Nahiyan Khushbu

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Summary

My expertise lies at the intersection of **Deep Learning** and **Computer Vision**, with a specialization in innovative **Medical AI** methodologies. I have successfully designed and implemented novel **hybrid CNN-Transformer models** capable of processing both 2D and 3D imaging data for high-impact applications such as early Parkinson's Disease recognition. I excel at bridging theoretical AI research (architecture design, attention mechanisms) with practical engineering skills (Python, PyTorch, Docker), ensuring my contributions are both mathematically sound and ready for real-world deployment. I seek to leverage these skills to conduct foundational research in AI-assisted diagnostics.

Education

2022 - 2025 Master of Science (Computer Science and Technology) at **Southwest Jiaotong University, China** (GPA: 3.38/4.0)
2017 - 2021 Bachelor of Engineering (Software Engineering) at **Sichuan University** (GPA: 3.02/4.0)

Publications

1. Khushbu A N, Yang Z, Liu Y, Zhang X. **Parkinson's Disease Recognition Using Hybrid CNN-Transformer Model** [C], in *2024 4th International Conference on Industrial Automation, Robotics and Control Engineering (IARCE 2024)*, IEEE, 2024, pp. 307-311.
DOI: 10.1109/IARCE64300.2024.00064.

   Award: Best Paper Award at IARCE-2024.

2. Khushbu A N, Zhang X, et al. **HybridTransMedFormer: A Dual-Path 3D MRI Fusion Framework for Early Parkinson's Disease Recognition** [J], *Artificial Intelligence in Medicine*. (Under Review, Expected 2026).

Projects/Research Experience

Master's Thesis: Hybrid Deep Learning Frameworks for Early Parkinson's Disease Recognition: Bridging 2D and 3D MRI Innovations 05/2025
Developed a two-tier diagnostic pipeline using structural MRI: a lightweight **Hybrid CNN-Transformer (80.51% accuracy)** for 2D rapid pre-screening, and the robust **Hybrid-TransMedFormer (92.34% accuracy)** 3D dual-path fusion architecture for confirmatory diagnosis. This system is projected to reduce diagnostic delays by 12–18 months for prodromal patients.

Bachelor's Thesis: Bird Species Identification using Deep Neural Networks 05/2021
Developed an image recognition system to identify bird species, providing a helpful tool for bird-watchers. The solution leveraged **Transfer Learning** using pre-trained **VGG-16** and **ResNet18** architectures on the Caltech-UCSD Birds 200 (CUB-200-2011) dataset. The pipeline included image pre-processing, data augmentation, and utilized **PyTorch** and **NumPy** for model training and comparison.

GlobalfaceNet- Emotion Detection from Facial Expressions (Unimodal Emotion Recognition) 04/2025

Designed the novel **GlobalFaceNet** architecture, the first **CNN-Transformer hybrid** for unimodal emotion recognition. Employs **dual-pathway processing** to integrate fine-grained facial details with global spatial context, achieving **85.0% accuracy** on the RAF-DB dataset with fast **3.2ms inference time**.

TransFormerNet- Emotion Detection from Facial Expressions (Multimodal Emotion Recognition) 06/2025

Developed **TransFormerNet**, a **multimodal framework** leveraging cross-modal attention fusion to combine visual and comprehensive 39-dimensional auditory features. Achieved a transformative **99.7% accuracy** and **AUC 1.000** on balanced datasets, demonstrating superior performance by integrating complementary cross-modal information.

Image Processing detecting Fractured Bones 07/2020

Developed an image processing pipeline to analyze X-ray images, applying filtering and edge detection techniques to automatically highlight and detect potential bone fractures.

AI Based e-Commerce Website (Intern Project) 01/2021

Contributed to the development of an e-commerce platform incorporating an AI-driven recommendation engine based on user history and real-time browsing behavior.

COVID-19 Facemask Detection with Deep Learning 06/2021

Trained a Convolutional Neural Network (CNN) model to accurately identify the presence or absence of a facemask on individuals in images and video streams.

Technical Skills

Category	Frameworks/Languages	Libraries/Concepts
AI/Deep Learning	Python, Java	PyTorch, TensorFlow, Keras, **Attention Mechanisms,**
Medical/Vision	NiBabel, Scikit-image	OpenCV, Grad-CAM, **3D Data Augmentation,**
Deployment/Tools	**FastAPI / Flask**, SQL	**Docker**, **Git / GitHub**, Deployment,
Data Analysis		NumPy, Pandas, Matplotlib, Seaborn.

Work Experience

Software Requirements Research and Publisher, SoftFlames Ventures LLC 01/11/2025 – Present

City: Hamtramck, Michigan — Country: USA

Software Engineer Intern, Chengdu Suncape Data Co., Ltd 01/09/2020 – 31/03/2021

City: Chengdu — Country: China